BRENDAN JOYCE

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SUMMARY

Data analyst with 100+ hours of graduate coursework in R and reproducible research workflows. Comfortable taking raw, multi-source data through cleaning, transformation, and visualization in the tidyverse. Project work spans security and political analysis, including Janes Intelligence Review, the Correlates of War Arms & Technology dataset, and longitudinal environmental data.

TECHNICAL SKILLS

- **Languages & Tools:** R (tidyverse, dplyr, tidyr, ggplot2, lubridate, patchwork, ggtext, knitr/kable), Quarto, R Markdown, RStudio, Git/GitHub, Excel, basic SQL.
- **Data Wrangling:** multi-source joins, pivots, regex cleaning, custom functions, HTML web scraping.
- **Visualization:** ggplot2 (bar, scatter, histogram, ribbon/area), multi-panel layouts with patchwork, in-plot annotation with ggtext, formatted tables with kable, RevealJS slide decks.
- **Reproducible Research:** Quarto and R Markdown notebooks, version control with Git/GitHub, documented code structured for replication.

SELECTED DATA ANALYSIS PROJECTS

International Response to the 2022 Russian Invasion of Ukraine

- Used Janes Intelligence Review data in a Quarto-based workflow to compare state-level diplomatic and material responses to the invasion.
- Joined and reshaped records from multiple sources using dplyr and tidyr, then aggregated to country-level summaries to compare allied, neutral, and adversarial responses.
- Built a set of ggplot2 charts and short written analysis framed for a security-policy audience.

Military Modernization & Technology Diffusion (Armed UAVs)

- Built side-by-side comparisons across weapon systems in ggplot2 to trace how armed UAV capabilities have spread between states over time.
- Combined multiple plots into a single multi-panel figure with patchwork and wrote a short interpretation alongside it.

Webscraping

- Wrote an R function using rvest to scrape U.S. senator profile data from the C-SPAN website, parsing HTML elements into a tidy data frame.
- Used CSS selectors and regex to pull out keywords and surface patterns across the scraped text.

Replication & Improvement of Published Political Science Visualizations

- Replicated Figure 2 from the 2021 Suppressing Black Votes case study on voter registration and the Louisiana Understanding Clause, then redesigned axes, color scales, and annotations to make the original finding easier to read.
- Recreated a ribbon plot of Trump political popularity to practice plotting uncertainty bands in ggplot2.

Longitudinal Analysis of Global Environmental Change

- Reshaped the Greenspace dataset from raw observations into tidy, analysis-ready tibbles and built time-series plots to track land-cover change across time.

EDUCATION

American University, School of International Service — Washington, DC

Master of Arts, International Affairs — Global Governance, Politics, and Security — Expected May 2026

- Coursework: SIS 750 Data Analysis, 100+ hours in R, tidyverse, reproducible workflows, visualization.